Problem Set for 31 March 2025

Exercise 1 For $U \xrightarrow{f} V \xrightarrow{g} U$ ($\dim < \infty$), show that

$$\dim \operatorname{im}(f) + \dim \operatorname{im}(g) \leq \dim \operatorname{im}(g \circ f) + \dim V. \tag{$\star$}$$

The equality holds if and only if $\ker(g) \subset \operatorname{im}(f)$.

- **(optional, but thought-provoking)** Can thee bethink of aught analogous propositions in matrix theory?

Exercise 2 Let $\mathbb{F}$ be arbitrary ground field with character $\neq 2$, i.e., $1+1 \neq 0$. Let $A, B \in \mathbb{F}^{2m \times 2m}$ be matrices, such that

$$A^2 = B^2 = I_{2m}, \quad AB + BA = O_{2m}.$$

Show that there exists $P \in \mathrm{GL}_{2m}(\mathbb{F})$ such that

$$PAP^{-1} = \begin{pmatrix} I_m & O \\ O & -I_m \end{pmatrix}, \quad PBP^{-1} = \begin{pmatrix} O & I_m \\ I_m & O \end{pmatrix}.$$

- **(optional, but very easy)** For what kind of $z \in \mathbb{C}$, one can find $A, B \in \mathbb{C}^{2025 \times 2025}$ such that $AB = z \cdot BA \neq O$?
- **(optional, but very easy)** Find non-examples when $\mathbb{F}$ has characteristic 2, i.e., $1+1=0$.

Exercise 3 Let $V = \mathrm{Hom}_{\mathrm{Sets}}(\mathbb{R}, \mathbb{R})$ be the set of continuous functions over $\mathbb{R}$, which also an $\mathbb{R}$-vector spaces. We know that for any $x \in \mathbb{R}$, the map

$$\delta_x : \mathrm{Hom}_{\mathrm{Sets}}(\mathbb{R}, \mathbb{R}) \to \mathbb{R}, \quad f \mapsto f(x)$$

is an surjection.

1. Show that $\bigcap_{x \in \mathbb{R}} \ker \delta_x = 0$ (in few words).
2. Moreover, show that $\{f_i\}_{i=1}^n \subset V$ is linearly independent **if and only if** there exists $\{x_i\}_{i=1}^n \subset \mathbb{R}$ such that

$$\det(A_{n \times n}) \neq 0, \quad a_{i,j} = f_i(x_j).$$

Exercise 4 (optional) (2023 年上海交通大学考研) 设 U, V, W 是给定域上的线性空间. $U \xrightarrow{\alpha} V$ 与 $V \xrightarrow{\beta} W$ 是线性映射且满足 $U \xrightarrow{\beta\circ\alpha} W$ 是零映射. 若对任何一个线性空间 X 和线性映射 $V \xrightarrow{f} X$ 使得 $U \xrightarrow{f\circ\alpha} X$ 为零, 都存在一个唯一的线性映射 $W \xrightarrow{\mu} X$ 使得 $f = \mu \circ \beta$, 证明

1. β 是满射 (in few sentences).
    - 提示: β 是满射当且仅当对任意 X, 线性映射 $\mathrm{Hom}_{\mathbb{F}}(W, X) \xrightarrow{-\circ\beta} \mathrm{Hom}_{\mathbb{F}}(V, X)$ 总是单射.
2. β 诱导了线性同构 $W \simeq V/\mathrm{im}(\alpha)$. 换言之, 证明 β 是满的, 且 $\ker(\beta) = \mathrm{im}(\alpha)$.